MEMORY SCENT RECIPER

ABSTRACT

This project presents an innovative Generative AI-based system that transforms human memory descriptions into structured scent formulations. The system leverages a fine-tuned GPT-2 model to analyze textual inputs and generate fragrance components such as top notes, middle notes, base notes, and corresponding chemical molecules.

Human memories are often closely associated with sensory experiences, especially smell. However, converting these abstract and emotional memories into structured fragrance compositions is a complex task. This project addresses that challenge by using artificial intelligence to simulate human-like interpretation of sensory input. The system demonstrates how AI can go beyond traditional applications and contribute to creative domains such as perfume design, sensory computing, and personalized user experiences.

PROBLEM STATEMENT

Smell is one of the most powerful human senses, strongly connected to memory and emotion. A single fragrance can instantly remind us of a specific place, person, or moment in our life. However, recreating these memories in the form of actual scents is a complex and challenging process.

- No automated system to convert human memories into fragrance compositions
- Traditional perfume creation is time-consuming and requires expert knowledge
- Difficulty in translating abstract sensory experiences into structured data
- Lack of intelligent systems that combine creativity with chemical representation
- Need for an AI-driven approach to simplify and automate scent generation

With the advancement of Artificial Intelligence, especially in the field of Natural Language Processing, it is now possible to interpret human language and generate meaningful outputs. This project explores how AI can be used creatively to transform memory descriptions into structured scent formulations.

PROPOSED SOLUTION

To address the identified problem, we propose a Generative AI-based system that interprets memory descriptions and generates corresponding scent formulations. The system is designed to understand textual input and produce structured fragrance outputs in a meaningful and consistent format.

- Develop a system using Generative AI to interpret memory descriptions
- Use a trained language model to generate scent formulations
- Provide structured output including:
    - Top notes (initial impression)
    - Middle notes (core fragrance)
    - Base notes (long-lasting effect)
    - Chemical molecules
- Enhance output quality using dataset training and rule-based filtering
- Provide a user-friendly interface for interaction

The solution combines dataset-driven learning with rule-based refinement to improve the quality and reliability of generated results. Additionally, a simple and interactive user interface is provided to enhance usability.

END USERS

- Perfume designers and fragrance developers
- Researchers in Artificial Intelligence and sensory computing
- Creative professionals (artists, writers, designers)
- Students and academic researchers
- General users interested in personalized scent experiences
- Useful for creating signature scents for products, brands, or environments based on specific themes.
- Can be integrated with smart devices (like scent diffusers) for automated fragrance generation based on user input

IMPLEMENTATION OF GEN AI

The system is implemented using advanced Natural Language Processing techniques and Generative AI models. The GPT-2 model is used as the base architecture and is fine-tuned using both custom-created and real-world perfume datasets.

Data preprocessing is performed to structure the dataset into a consistent format suitable for training. Tokenization techniques are applied to convert textual data into machine-readable format. The model is trained using the Hugging Face Transformers library, which provides efficient tools for training and inference.

To improve output quality, prompt engineering and rule-based filtering techniques are applied. These ensure that the generated scent formulations are structured, meaningful, and relevant to the input memory.

The system uses Generative AI to convert a user's memory description into a structured scent formulation. It mimics human understanding of language and generates meaningful fragrance components based on learned patterns from data.

HOW IT WORKS IN YOUR SYSTEM

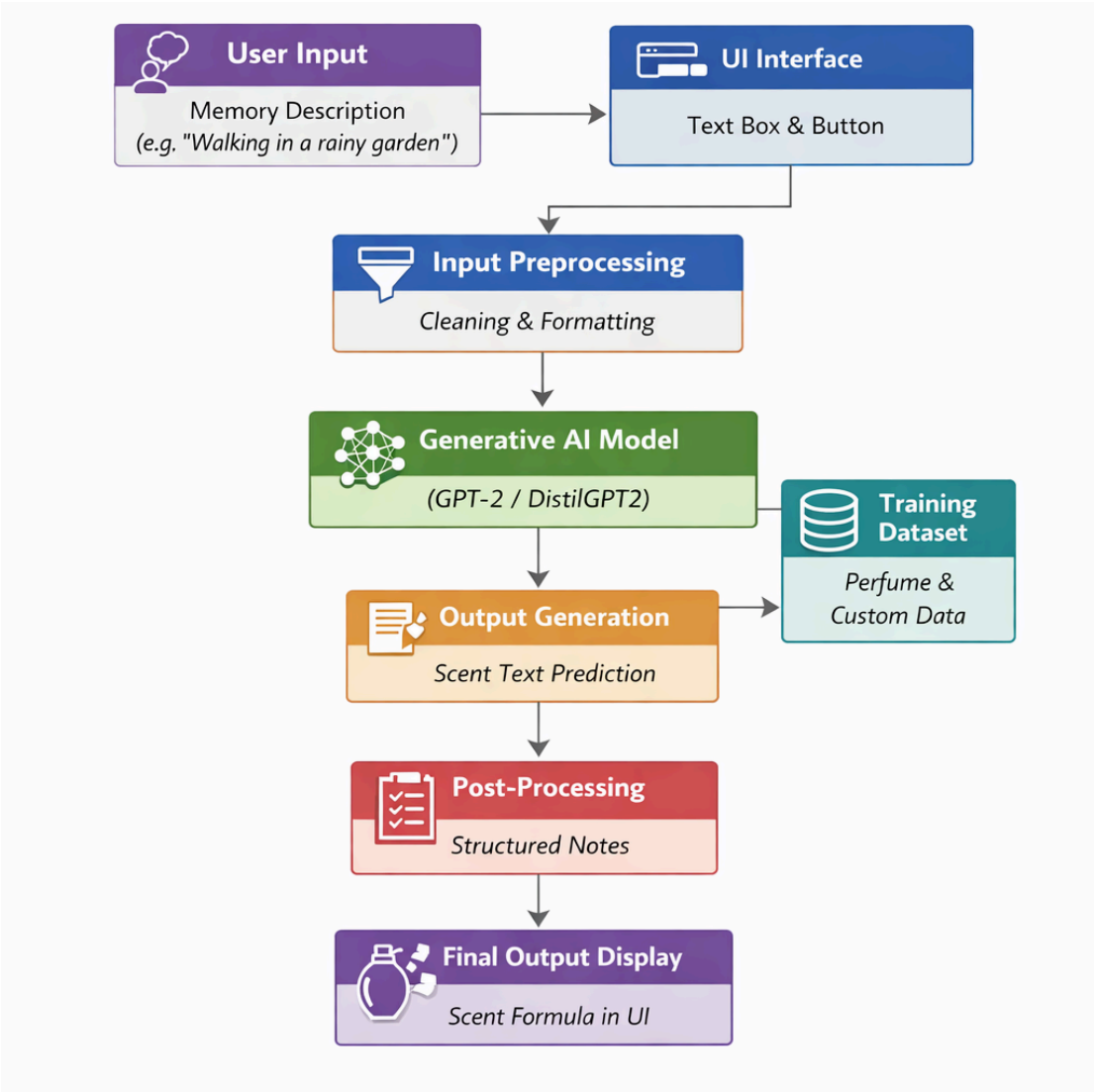
Step 1: User enters a memory description through UI. The user provides a memory description (e.g., "walking in a rainy garden").

Step 2: The system preprocesses the text by cleaning and formatting it for the model.
Step 3: The trained AI model generates scent-related text. A fine-tuned GPT-2 model generates scent-related text based on the input.It predicts the most relevant sequence of words representing fragrance notes.

Step-4: The model has been trained on perfume datasets containing descriptions and scent notes, enabling it to understand relationships between memory and smell.
Step 4: Post-processing ensures structured output (Top, Middle, Base, Molecules). The system cleans and filters the output to ensure clarity, consistency, and proper structure.
Step 5: Final result is displayed in a user-friendly interface

SYSTEM ARCHITECTURE

The proposed system follows a structured pipeline starting from user input to final scent generation. The architecture includes input preprocessing, a Generative AI model, and output post-processing.



GITHUB REPOSITORY LINK

The complete project, including implementation and documentation, is available on GitHub.

https://github.com/VARSHINI-S-30/memory-scent-reciper.git

FUTURE ENHANCEMENTS

In the future, this system can be enhanced by increasing the size and quality of the dataset to improve the accuracy and realism of generated scent formulations. Increase dataset size (5000+ entries) to improve accuracy and generate more realistic scent formulations. The current model can be upgraded to more advanced transformer architectures beyond GPT-2, which will provide better understanding of complex memory descriptions and more creative outputs. Additionally, integrating real-world chemical databases can help in generating scientifically accurate molecular compositions, making the system more useful for practical applications. Convert the system into a full-fledged application for real-world usage. The system can also be extended to support voice input, allowing users to describe their memories more naturally. Deploying the system as a mobile or web application will improve accessibility, while integration with smart devices like scent diffusers can enable real-time fragrance generation. Personalization features based on user preferences and past interactions can further enhance the overall user experience.

CONCLUSION

In conclusion, this project demonstrates the effective use of Generative AI in transforming human memory descriptions into structured scent formulations. By utilizing techniques from Natural Language Processing and a fine-tuned GPT-2 model, the system is capable of generating meaningful fragrance notes and chemical representations from textual input. This project highlights how artificial intelligence can extend beyond traditional problem-solving and contribute to creative and sensory domains. It provides a strong foundation for future developments in personalized fragrance design, intelligent systems, and human-centered AI applications. This work highlights the potential of artificial intelligence to move beyond traditional computational tasks and contribute to creative and sensory domains. The project not only provides a novel approach to scent generation but also opens new possibilities for innovation in personalized experiences, perfume design, and intelligent human-computer interaction.